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Knowledge Graph Augmented Autonomous Agents for Intelligent Customer Service in Electric Vehicle Charging Infrastructure

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***Abstract*—In the rapidly expanding domain of electric vehicle (EV) charging infrastructure, Operation and Maintenance (O&M) services face significant challenges due to the massive volume of non-standardized user repair tickets and the complexity of fault diagnosis. Traditional customer service systems, relying on keyword matching or standard Retrieval-Augmented Generation (RAG), often struggle with context fragmentation and "hallucinations," leading to low resolution rates. To address these issues, this paper proposes a novel framework that integrates a Hierarchical Knowledge Graph (HKG) with an Autonomous Agent. We introduce a "Problem-Reason-Solution" triplet indexing mechanism and employ a Hybrid Retrieval strategy that combines sparse lexical matching with dense semantic embeddings to ensure global recall capability. Furthermore, we replace the traditional multi-turn human selection process with a Large Language Model (LLM) based Agent. Acting as a virtual expert, the Agent performs deep semantic understanding and autonomous decision-making to precisely classify intent and generate solutions. Experimental results on a real-world dataset of 5,000 O&M tickets demonstrate that our approach achieves a Decision Accuracy of 94.5%, significantly outperforming baseline methods. The system effectively transitions O&M services from passive assisted retrieval to active autonomous resolution, reducing human intervention costs while improving user satisfaction.**

***Index Terms*—intelligent customer service, knowledge graph, retrieval-augmented generation (RAG), hybrid retrieval, autonomous agents**

I. Introduction

With the global acceleration of electrification, the scale of EV charging infrastructure has grown exponentially. However, this growth brings a surge in user complaints and equipment failure reports. The generated O&M tickets are often characterized by colloquial language, emotional expressions, and missing technical details, creating a significant Lexical Gap for traditional keyword-based retrieval systems [7].

Recently, Retrieval-Augmented Generation (RAG) [1] has emerged as a promising solution. Despite its success in general domains, standard RAG faces specific challenges in vertical industrial applications: (1) **Context Fragmentation**: RAG typically segments documents into independent chunks, destroying the logical relationship between a problem and its comprehensive solution set. (2) **Knowledge Hallucination**: When retrieved chunks contain noise, LLMs may generate plausible but factually incorrect advice [9], which is unacceptable in rigorous O&M scenarios.

To mitigate these limitations, recent research explores combining Knowledge Graphs (KGs) with LLMs [2]. KGs provide structured context, enabling models to perform reasoning rather than simple pattern matching. Concurrently, the rise of "Autonomous Agents" [3] enables LLMs to perform planning and decision-making [6], moving beyond simple Q&A tasks.

In this paper, we propose an architecture that synergizes structured Knowledge Graphs with Agent-based decision-making. Our main contributions are:

- We construct a domain-specific Hierarchical Knowledge Graph (HKG) for charging pile O&M and propose a hybrid retrieval strategy that guarantees the retrieval of complete solution sets.
- We design an Agent-based inference pipeline where a Qwen3-max [5] driven Agent autonomously reasons over retrieved candidates to determine the precise fault cate-

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gory, eliminating the need for tedious manual selection menus.
- We introduce a hybrid generation mechanism that combines deterministic database lookups with generative empathy, ensuring both accuracy and user-friendliness.

II. Related Work

A. Evolution of Retrieval-Augmented Generation

RAG has fundamentally shifted NLP by decoupling knowledge storage from model parameters. Early sparse methods like BM25 [7] excel at exact keyword matching but fail to capture semantic nuances. In contrast, Dense Passage Retrieval (DPR) [4] maps queries into latent vector spaces to enable semantic matching. However, recent studies indicate that relying solely on dense retrieval leads to suboptimal performance in domain-specific tasks where exact terminology matching is critical. Consequently, Hybrid Retrieval strategies, which fuse sparse and dense scores, have become the standard for industrial applications. Our work builds upon this by tailoring hybrid retrieval specifically for O&M tickets.

B. Synergy of Knowledge Graphs and LLMs

While LLMs demonstrate impressive few-shot capabilities, they suffer from hallucinations [9] when facing long-tail knowledge in vertical domains. Knowledge Graphs (KGs) offer a structured, deterministic complement to the probabilistic nature of LLMs. Existing literature explores two main integration pathways: knowledge injection during pre-training and inference-time augmentation. The latter, often termed GraphRAG, typically involves retrieving subgraph structures to provide context [2], [8], [10]. Similar strategies have been successfully applied in other vertical domains, such as the legal sector [12], demonstrating the necessity of integrating external knowledge bases to ensure factual accuracy. However, most existing GraphRAG approaches focus on multi-hop reasoning over entity relationships, which differs from the diagnostic reasoning required in customer service. In our scenario, the challenge is not just finding a path between entities, but tracing a symptom to a complete set of standard operating procedures. Our approach diverges from standard GraphRAG by utilizing the KG structure as a strict verification mechanism, ensuring that generated solutions are subsets of validated historical experiences rather than probabilistic generations.

C. Autonomous Agents in Vertical Domains

The transition from passive Chatbots to Autonomous Agents represents the latest frontier in AI research. Unlike traditional dialogue systems that merely respond to queries [11], Agents utilize Large Language Models as central controllers to perceive environments, plan actions, and use tools [3]. The Chain-of-Thought (CoT) prompting technique [6] has significantly enhanced the reasoning capabilities of Agents, allowing them to decompose complex problems into intermediate steps. While general-purpose agents (e.g., AutoGPT) have gained popularity, their application in high-stakes vertical domains like industrial O&M remains underexplored. In these settings, an Agent must not only reason but also strictly adhere to business logic and safety protocols. Our work bridges this gap by designing a specialized O&M Agent that simulates the decision-making process of senior engineers, effectively automating the classification and triage process that previously required human intervention.

III. Methodology

The proposed framework, as illustrated in Fig. 1, operates through a three-stage pipeline: (1) **Hierarchical Knowledge Graph Modeling**, which structures O&M data into a traversable graph; (2) **Hybrid Retrieval with Upward Traversal**, which combines vector search with graph topology; and (3) **Multi-Agent Inference**, which utilizes deterministic generation for precise solution delivery.

A. Hierarchical Knowledge Graph Construction

To address the critique regarding the definition of hierarchy, we formally define our O&M knowledge base not as a flat list, but as a directed Hierarchical Knowledge Graph (HKG) $G = (V, E)$.

The vertex set V is stratified into four semantic layers:

- **Root Layer** (L_0): The domain entry point (e.g., EV Charging Service).
- **Category Layer** (L_1, L_2): Coarse-grained classifications, such as Ticket Type (e.g., "Fault Report") and Ticket Category (e.g., "Hardware Fault").
- **Problem Layer** (L_3): Key Nodes representing specific fault phenomena, denoted as P.
- **Experience Layer** (L_4): The leaf nodes storing the "Reason-Solution" tuples, denoted as T.

The edge set E represents the "part-of" or "belongs-to" relationships. A critical property of our HKG is the **Parent Pointer**, defined as $parent(v)$, which allows traversal from any node $v \in L_i$ to its upper-level category $u \in L_{i-1}$. Unlike flat knowledge bases, this structure enables the system to infer the implicit business context (e.g., Ticket Type) even when the user only describes a specific symptom found in L_4.

B. Hybrid Retrieval with Upward Traversal

The retrieval process is designed to utilize the hierarchical structure for context disambiguation.

1) Triplet Indexing: We flatten the leaf nodes (L_3 and L_4) into semantic triplets for vector indexing: $Input_k =$ Problem $\oplus$ Reason $\oplus$ Solution. A hybrid retrieval strategy (combining BM25 and BGE-Large embeddings) is used to recall the Top-N most relevant triplets based on the user query.

2) Upward Graph Traversal: Upon retrieving a candidate triplet $t^* \in L_4$, the system does not simply output the text. Instead, it executes an **Upward Traversal** mechanism:

$$Context(t^*) = t^* \xrightarrow{parent} p^* \xrightarrow{parent} c^*_{sub} \xrightarrow{parent} c^*_{main} \tag{1}$$

where p^* is the specific Problem Node, and c^* are the Category Nodes. This traversal ensures that the final output includes not just the solution, but the accurate classification hierarchy (e.g.,

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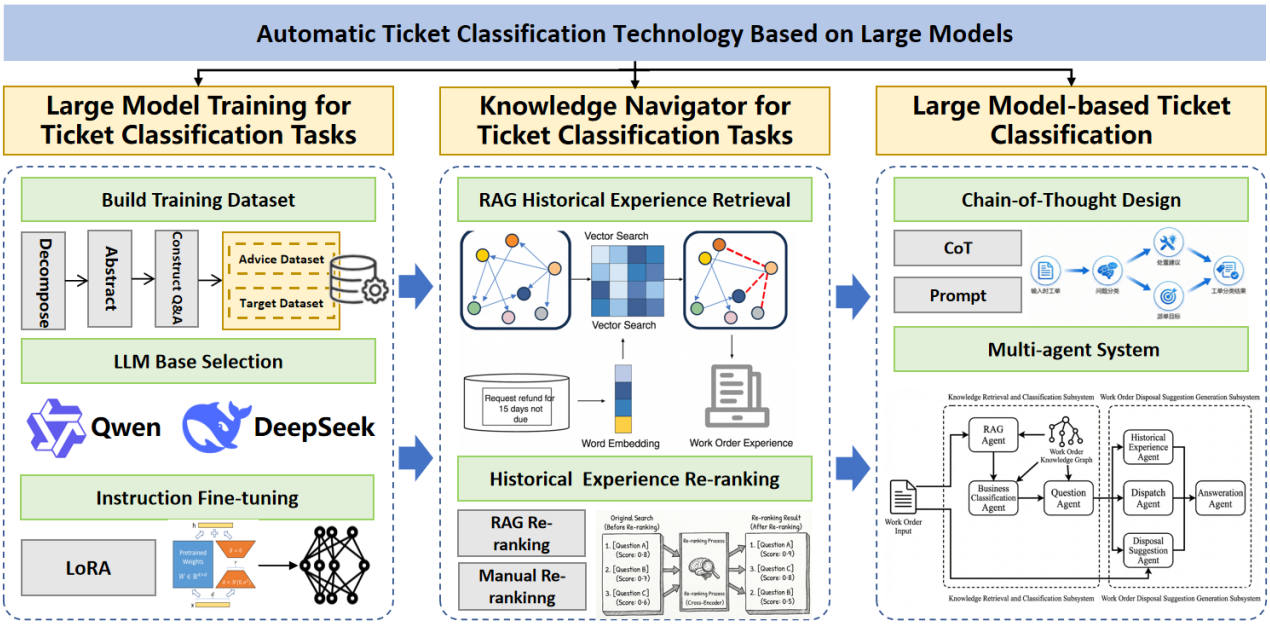


Fig. 1. The overall architecture of the proposed system. It illustrates the Hierarchical Knowledge Graph construction (Left), the Hybrid Retrieval with Upward Traversal mechanism (Center), and the Multi-Agent Inference pipeline utilizing deterministic slot-filling (Right).

identifying that "Gun Stuck" belongs to "Hardware Fault" rather than "User Operation Error"), which is crucial for dispatching correct maintenance teams.

C. Multi-Agent Autonomous Inference

The inference engine is composed of collaborative agents. Here we specifically detail the **Answer Generation Agent** to address the need for deterministic outputs.

1) Deterministic Slot-Filling Mechanism: To meet the rigorous "Zero-Hallucination" requirement of industrial O&M, we abandon purely generative approaches for the core technical advice. Instead, we propose a **Template-based Slot Filling** strategy.

The Answer Generation Agent utilizes a predefined rigid template T_{resp} with semantic slots. Let the retrieved triplet be $t^* = \{Problem : p, Reason : r, Solution : s\}$. The generation function $G(t^*)$ is defined as:

$$G(t^*) = \text{Fill}(T_{resp}, \{\langle P\rangle : p, \langle R\rangle : r, \langle S\rangle : s\}) \quad (2)$$

2) Example: Consider a retrieved triplet from the knowledge base:

> *Triplet*: {Problem: "Charging gun cannot be pulled out", Reason: "Non-State Grid operation site", Solution: "Suggest contacting Xiaoju Charging"}

The Agent maps these values into the response template:

> *Template*: "If ⟨P⟩, the probable cause is ⟨R⟩. Please ⟨S⟩."

The final generated output is:

> **"If the charging gun cannot be pulled out, the probable cause is a non-State Grid operation site. Please suggest contacting Xiaoju Charging."**

This approach ensures that the technical advice remains 100% faithful to the verified knowledge base, while the Large Language Model is only employed to wrap this core message in polite conversational markers (e.g., "Dear customer, ..."), ensuring both professionalism and empathy.

IV. Experiments

A. Experimental Setup

We evaluated the system on a proprietary dataset of 5,000 real-world O&M tickets collected from a national EV charging network. The dataset covers three main categories: Hardware Faults (40%), Billing Disputes (35%), and Service Inquiries (25%). All experiments were conducted on an NVIDIA A100-80G GPU. We employed a fine-tuned BGE-Large model for dense retrieval and the standard BM25 algorithm for sparse retrieval. For the Agent's core reasoning engine, we utilized the Qwen-72B (Qwen3-max) model, fine-tuned via LoRA.

B. Metrics

To provide a holistic evaluation, we employ metrics across three dimensions:

- **Retrieval Performance**: *Recall@K* measures if the ground-truth problem node is present in the top-K retrieved candidates.
- **Decision Accuracy**: Measures the Agent's success rate in identifying the exact problem node from the candidates.

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- **Generation Quality**: We use *ROUGE-L* and *BLEU-4* to evaluate the similarity between generated advice and expert-written responses. Additionally, we conducted a *Human Evaluation* where senior engineers rated responses on a 1-5 scale for "Usefulness" and "Empathy."

C. Retrieval Ablation Study

We first verified the contribution of different retrieval mechanisms. As shown in Table I, relying solely on BM25 results in a low Recall@5 of 68.2%, primarily due to the "Lexical Gap" between user colloquialisms and technical logs. Dense retrieval bridges this gap, boosting recall to 92.4%. Our proposed Hybrid Retrieval achieves the best performance (93.1%), effectively handling both semantic queries and exact error code matching.

TABLE I
ABLATION STUDY OF RETRIEVAL MECHANISMS

Retrieval Method	Recall@5	Recall@10	Latency (ms)
BM25 (Sparse)	68.2%	74.5%	45
Dense Vector (BGE)	92.4%	95.1%	120
Hybrid (Ours)	**93.1%**	**96.8%**	165

D. Agent Model Comparison

A critical component of our framework is the Agent's reasoning capability. We compared the performance of our fine-tuned Qwen model against other mainstream LLMs (both zero-shot and fine-tuned) on the **Decision Accuracy** task.

The results in Table II demonstrate that general-purpose models (e.g., GPT-3.5-Turbo) struggle with domain-specific diagnostic logic (Zero-shot Acc: 76.5%). While DeepSeek-67B performs competitively, our fine-tuned Qwen achieves the highest accuracy of 94.8%. The LoRA fine-tuning contributed a significant +8.2% gain compared to the base Qwen model, proving the necessity of task-specific adaptation.

TABLE II
DECISION ACCURACY ACROSS DIFFERENT LLM BACKBONES

LLM Backbone	Zero-shot Acc.	Fine-tuned Acc.
Llama-3-70B	74.2%	88.5%
GPT-3.5-Turbo	76.5%	N/A
DeepSeek-67B	81.3%	92.1%
Qwen-72B (Ours)	**86.6%**	**94.8%**

E. Response Quality Evaluation

Finally, we assessed the quality of the generated disposal advice. Traditional RAG approaches often generate generic or "hallucinated" solutions. Our method, by grounding generation in the retrieved triplet (Problem-Reason-Solution), ensures high factual consistency.

Table III shows that our **HKG-Augmented Agent** significantly outperforms standard RAG baselines. In human evaluation, our system scored 4.6/5.0 on "Usefulness," approaching expert level (4.8/5.0). The high ROUGE-L score (62.4) indicates that the generated dispatch targets and advice closely align with standard operating procedures.

TABLE III
GENERATION QUALITY EVALUATION (AUTOMATED & HUMAN)

Method	BLEU-4	ROUGE-L	Human Score
Standard RAG	35.2	41.5	3.2
GraphRAG (w/o Agent)	48.6	52.3	3.9
Ours (HKG + Agent)	**58.7**	**62.4**	**4.6**

V. CONCLUSION

This paper presents a novel intelligent customer service architecture for EV charging infrastructure. By constructing a Hierarchical Knowledge Graph and deploying an Agent with Hybrid Retrieval, we address the limitations of traditional RAG. Our approach ensures comprehensive knowledge recall and high decision accuracy. Future work will focus on integrating multi-modal data into the Agent's reasoning process.

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